

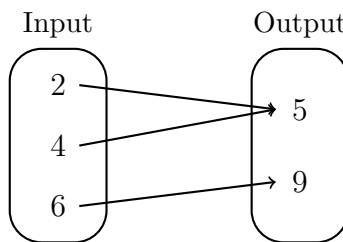
# Deciding Whether a Relation Is a Function

*Deciding from mapping diagrams, from sets and tables, and from graphs with the vertical line test*

**Directions.** A relation is a **function** when every input has exactly *one* output. Use the same two-step count every time:

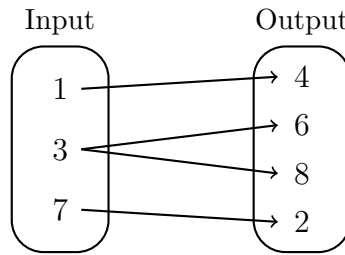
- Count the **pairs** (arrows, rows, or plotted points).
- Count the **different inputs** (left-hand values, or first coordinates).
- If the two counts are **equal**, no input was used twice, so the relation is a function. If the number of different inputs is **less** than the number of pairs, some input has two outputs and it is not a function.
- On a graph the same test is the **vertical line test**: if any vertical line meets the graph more than once, that input has two outputs.
- Outputs are allowed to repeat. Only *inputs* may not.

**1.** (a) Which output does the input 4 send to? (b) Count the arrows and the different inputs, write the two counts with  $<$ ,  $>$  or  $=$  between them, and say whether this is a function.



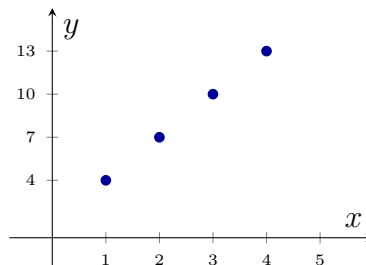
Answer: \_\_\_\_\_

**2.** Count the arrows and the different inputs in this diagram, write the two counts with  $<$ ,  $>$  or  $=$  between them, and say whether this relation is a function. If it is not, name the input that spoils it.



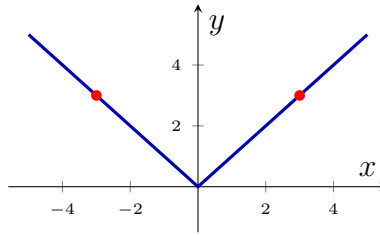
Answer: \_\_\_\_\_

**3.** The relation  $\{(1, 4), (2, 7), (3, 10), (4, 13)\}$  is plotted below. **(a)** Count the points and the different first coordinates and compare the two counts; is it a function? **(b)** Every pair fits one rule. Write the rule in the form  $y = 3x + 1$  and use it to give the output when the input is 5.



Answer: \_\_\_\_\_

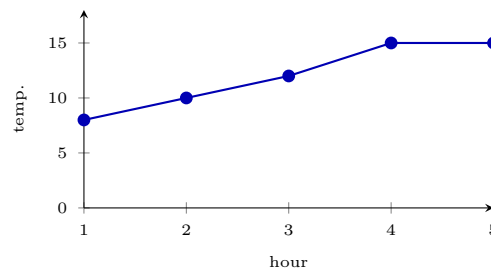
4. The graph below is  $y = |x|$ . (a) Read the output at the input  $-3$ , then at the input 3. (b) Does this graph pass the vertical line test? (c) Two different inputs gave the same output. Explain in a sentence why that does *not* stop it being a function, while one input with two outputs would.



Answer: \_\_\_\_\_

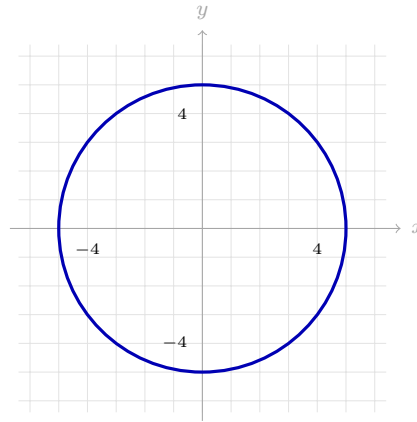
5. A thermometer was read once an hour. (a) Give the temperature at hour 3. (b) Count the readings and the different hours, compare the two counts, and say whether “hour  $\rightarrow$  temperature” is a function.

| Hour  | 1 | 2  | 3 | 4  | 5  |
|-------|---|----|---|----|----|
| Temp. | 8 | 10 |   | 15 | 15 |



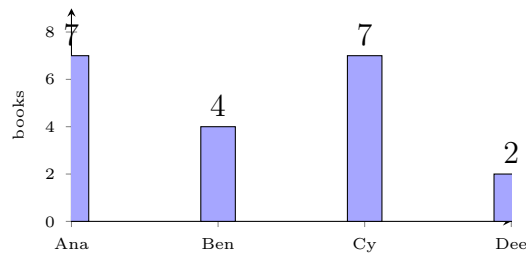
Answer: \_\_\_\_\_

**6.** The circle below has radius 5. **(a)** The vertical line through the input 3 meets it twice; give the upper of the two outputs. **(b)** A function allows one output per input — write that count and the count this graph gives with  $<$ ,  $>$  or  $=$  between them. **(c)** State the vertical line test verdict and justify it by naming a vertical line that meets the graph twice.



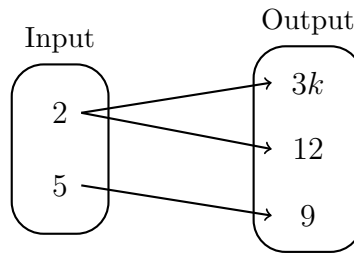
Answer: \_\_\_\_\_

**7.** The chart shows how many books four students read. **(a)** How many did Ben read? **(b)** What is the largest total? **(c)** Now reverse the relation, so the input is the number of books and the output is the reader. Count the arrows and the different inputs of the reversed relation, compare them, and say whether the reversed relation is a function.



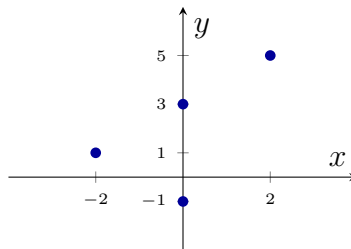
Answer: \_\_\_\_\_

8. In this diagram the input 2 still has two arrows leaving it, so the only way the diagram can show a function is if both arrows land on the same output. Set the two outputs equal and solve for  $k$ .



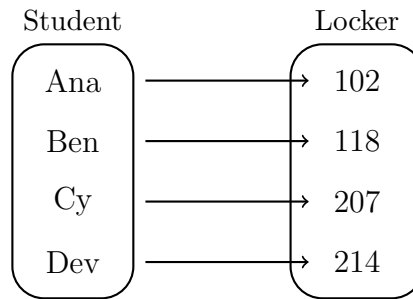
Answer: \_\_\_\_\_

9. Four points are plotted below. **(a)** Count the points and the different first coordinates, compare the counts, and say whether the relation is a function. **(b)** Three of the four points lie on the line  $y = x + 3$ . Check that claim for the point whose input is 2, and say which single point you would delete to repair the relation.



Answer: \_\_\_\_\_

**10.** Each student is given one locker. **(a)** Which locker does Dev get? **(b)** Count the arrows and the different students, compare the counts, and say whether “student  $\rightarrow$  locker” is a function. **(c)** Reverse the arrows so lockers are the inputs. Explain why that is also a function here, and describe the one change to the diagram that would stop it being one.



Answer: \_\_\_\_\_